Action Potential Propagation in Branched Neuronal Cables

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Abstract

Neuronal branching couples membrane excitability to the electrical load imposed by connected cables. This report develops a conservative numerical model of action-potential propagation on a tree of Hodgkin–Huxley cables. Membrane area is assigned consistently at junctions and endpoints, voltage is advanced by a staggered Crank–Nicolson method, and the tree structure permits a linear-time voltage solve. Analytic passive-cable and independent active-membrane benchmarks exhibit approximately second-order convergence in the tested smooth cases. In an idealized bifurcation, a fixed suprathreshold pulse reaches both daughter endpoints for all five tested diameter ratios. Matched endpoint stimulation on a uniform cable produces reflection-symmetric propagation, while additional membrane area at one endpoint changes the measured arrival times. These calculations provide a numerical framework for studying how geometry and stimulation shape propagation; cell-specific calibration and general propagation criteria remain open questions.

1 Introduction

An action potential propagates through a balance between regenerative membrane currents and axial current flowing into neighboring regions. At a branch point, the incoming wave must charge the membrane of multiple daughter cables. Cable diameter, length, membrane properties, and stimulus therefore influence both the waveform and its arrival time [1, 3].

The model considered here represents a neuron as a connected, acyclic network of cylindrical cables. Each cable is one-dimensional along its axis; branching specifies how cables exchange current. A drawing of the tree may be embedded in three dimensions, but this is a cable model rather than a three-dimensional extracellular-field calculation. The study develops a conservative solver and tests propagation through a bifurcation and along a cable in opposite directions.

The simulations use homogeneous Hodgkin–Huxley membrane kinetics and explicitly specified idealized geometry. Their dimensions and channel parameters are numerical model choices, not measurements of a specific neuron. This distinction permits clear tests of the equations while keeping biological interpretation within the evidence.

2 Cable and membrane equations

On a cable of constant radius r, let x denote axial distance and $V(x,t)$ the membrane potential. With specific membrane capacitance C and intracellular resistivity ρ,

$$C\frac{\partial V}{\partial t} + I_{\text{ion}}(V,m,h,n) = \frac{r}{2\rho}\frac{\partial^2 V}{\partial x^2}. \tag{1}$$

The ionic current density is

$$I_{\text{ion}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}}) + \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}) + g_{\text{L}}(V - E_{\text{L}}), \tag{2}$$

where m and h are sodium activation and inactivation gates and n is the potassium activation gate. Each gate satisfies

$$\dot{s} = \alpha_s(V)(1-s) - \beta_s(V)s, \qquad s \in \{m, h, n\}. \tag{3}$$

Outward ionic current is positive; injected current is positive inward. The axial current in the positive x direction is $i = -\pi r^2 \partial_x V/\rho$.

Table 1: Membrane parameters used throughout the active-cable experiments. Lengths are in cm, time in ms, voltage in mV, and total injected current in μA.

Parameter	Value	Unit
Specific capacitance C	1	μF cm^{-2}
Maximum conductances $(\bar{g}_{\mathrm{Na}}, \bar{g}_{\mathrm{K}}, g_{\mathrm{L}})$	$(120, 36, 0.3)$	mS cm^{-2}
Reversal potentials $(E_{\mathrm{Na}}, E_{\mathrm{K}}, E_{\mathrm{L}})$	$(45, -82, -59)$	mV
Axial resistivity ρ	0.0354	kΩ cm

For the voltage convention in Table 1, define $f(z) = z/(1-e^{-z})$ with $f(0) = 1$. The rate functions, in ms^{-1}, are

$$\begin{aligned}
\alpha_m &= f((V+45)/10), & \beta_m &= 4e^{-(V+70)/18}, \\
\alpha_h &= 0.07e^{-(V+70)/20}, & \beta_h &= (1+e^{-(V+40)/10})^{-1}, \\
\alpha_n &= 0.1f((V+60)/10), & \beta_n &= 0.125e^{-(V+70)/80}.
\end{aligned}$$

The removable singularities at $V = -45$ and -60 mV are evaluated using a local series and `expm1`. The active propagation experiments start at the uniform equilibrium $V_* \approx -69.89747$ mV and the gates are initialized to $s_\infty(V_*) = \alpha_s(V_*)/[\alpha_s(V_*) + \beta_s(V_*)]$.

Voltage is continuous at a junction. Direct each incident cable coordinate outward from the junction. For additional lumped membrane area A_0 and applied total current $I_0(t)$, conservation gives

$$I_0(t) = A_0[C\dot{V}_0 + I_{\mathrm{ion},0}] - \sum_{\ell} \frac{\pi r_\ell^2}{\rho} \partial_{x_\ell} V_\ell(0,t). \tag{4}$$

An unstimulated terminal without added lumped area has a sealed-end condition. An endpoint receiving a point current instead obeys the corresponding current balance. The root identifies the orientation used to store the tree; it represents a soma only when an explicit soma model is supplied.

3 Conservative numerical method

3.1 Spatial discretization

Divide each cable into segments, allowing a different mesh spacing on each branch. A segment e of length Δx_e and radius r_e has axial conductance

$$K_e = \frac{\pi r_e^2}{\rho \Delta x_e}. \tag{5}$$

Half of its lateral membrane area, $\pi r_e \Delta x_e$, belongs to each endpoint control volume. The membrane area at node i is therefore

$$A_i = A_i^{\mathrm{extra}} + \sum_{e \sim i} \pi r_e \Delta x_e. \tag{6}$$

This construction gives the correct half-segment area at a sealed terminal and combines contributions from every incident branch at a junction [4]. Additional physical membrane area is distinct from these mesh contributions. If membrane components differ, their ionic conductances and reversal-current terms are summed in proportion to their areas.

Let $M = \mathrm{diag}(CA_i)$ and let L be the weighted graph Laplacian with conductances K_e. The axial term is $(LV)_i = \sum_{j\sim i} K_{ij}(V_i - V_j)$. Every edge current appears with opposite signs at its two endpoints, so the discretization conserves axial current exactly at the algebraic level.

3.2 Time stepping and tree solution

Gates are stored at half time levels and voltage at integer time levels. With rates evaluated at V^k, the gate update is

$$s^{k+1/2} = \frac{[1 - \frac{1}{2}\Delta t(\alpha + \beta)]s^{k-1/2} + \Delta t\alpha}{1 + \frac{1}{2}\Delta t(\alpha + \beta)}. \tag{7}$$

For the first update, gates supplied at $t = 0$ are advanced by $\Delta t/2$ with the rates evaluated at V^0. This places them at the correct half time level.

Using these gates, write $I_{\mathrm{ion},i} = g_i V_i - b_i$, where

$$g_i = \bar{g}_{\mathrm{Na}} m_i^3 h_i + \bar{g}_{\mathrm{K}} n_i^4 + g_{\mathrm{L}},$$
$$b_i = \bar{g}_{\mathrm{Na}} m_i^3 h_i E_{\mathrm{Na}} + \bar{g}_{\mathrm{K}} n_i^4 E_{\mathrm{K}} + g_{\mathrm{L}} E_{\mathrm{L}}.$$

Set $G = \mathrm{diag}(A_i g_i)$ and $q_i = A_i b_i$. The voltage update is

$$[M + \tfrac{1}{2}\Delta t(L + G)]V^{k+1} = [M - \tfrac{1}{2}\Delta t(L + G)]V^k + \Delta t(q + \bar{I}^k), \tag{8}$$

where $\bar{I}^k = \Delta t^{-1} \int_{t_k}^{t_{k+1}} I(t)\, dt$. Exact integrals of the prescribed pulses preserve injected charge even when pulse edges fall between time steps.

Numbering parents before children permits elimination from leaves to root, followed by recovery from root to leaves. Because a tree has no cycles, elimination introduces no connections between separate branches, and the voltage solve takes $O(N)$ operations for N nodes [2, 4]. An independently assembled sparse linear solve provides a numerical reference. Cyclic and disconnected inputs are excluded.

3.3 Admissibility and time step selection

Crank–Nicolson does not by itself guarantee that nonlinear gate values remain in $[0, 1]$. For a gate with nonnegative rates, a sufficient condition for the update in Eq. 7 to map the entire interval into itself with nonnegative slope is

$$\Delta t\, [\alpha_s(V) + \beta_s(V)] \le 2. \tag{9}$$

Larger values can produce inadmissible gates. Runs are checked for this failure rather than silently clipping gates. Likewise, the voltage scheme can exhibit oscillatory stiff-mode behavior. The condition

$$\Delta t \le \min_i \frac{2CA_i}{\sum_{e\sim i} K_e + A_i g_i} \tag{10}$$

ensures nonnegative diagonal entries in the right-hand matrix of Eq. 8; it is a sufficient matrix-sign condition, not a necessary stability limit or an accuracy guarantee. Gate bounds, current residuals, and mesh and time refinement answer different questions and are evaluated separately.

4 Numerical verification

The verification suite covers membrane area, current conservation, heterogeneous junctions, tree topology, pulse charge, gate-rate singularities, equilibrium preservation, and agreement between tree and sparse solvers. Two independent reference problems assess discretization error.

For a passive, sealed cable of length $\ell = 1$ cm and radius $r = 0.0238$ cm, the initial condition is $V(x, 0) = E_{\mathrm{L}} + 5\cos(\pi x/\ell)$. Its exact solution is

$$V(x,t) = E_{\mathrm{L}} + 5\cos(\pi x/\ell)\exp\left[-\frac{g_{\mathrm{L}} + r\pi^2/(2\rho\ell^2)}{C}\,t\right]. \tag{11}$$

At $t = 0.1$ ms and $\Delta t = 0.0005$ ms, increasing the mesh from 10 to 20, 40, and 80 intervals reduces the maximum voltage error from 9.48×10^{-3} to 2.37×10^{-3}, 5.93×10^{-4}, and 1.48×10^{-4} mV. The corresponding observed orders are 1.998, 2.000, and 2.002.

For the active reference problem, voltage is spatially uniform, so axial current vanishes. The initial voltage is -55 mV, with gates equilibrated at V_*. A separate DOP853 integration of the four-variable membrane ODE, using relative tolerance 10^{-12} and absolute tolerance 10^{-13}, provides the reference voltage at 1 ms. With half-step gate initialization, time steps of 0.02, 0.01, 0.005, and 0.0025 ms give errors of 0.1130, 0.02787, 0.006937, and 0.001732 mV, respectively. Observed orders are 2.019, 2.006, and 2.002.

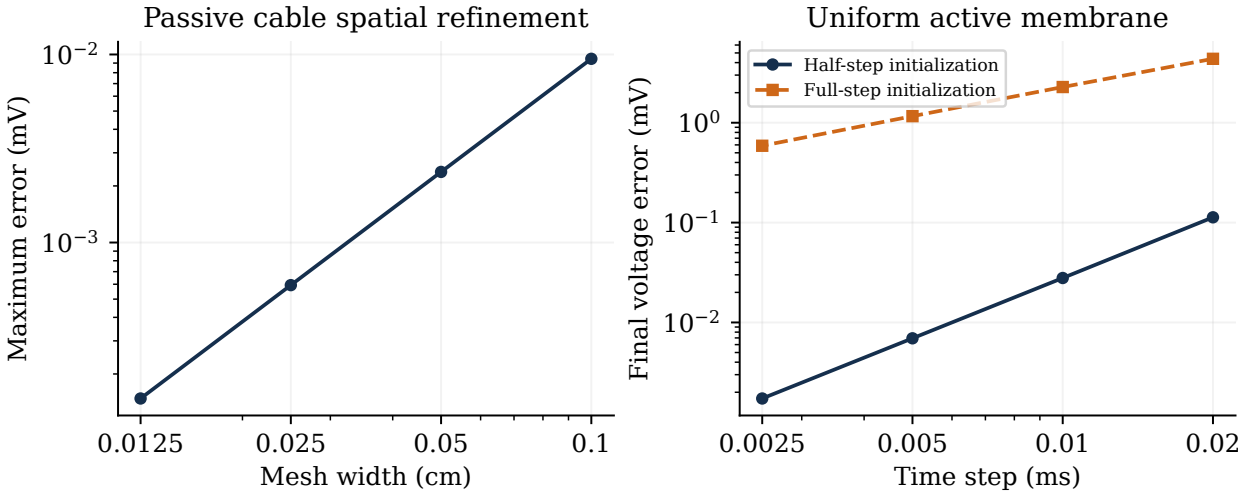


Figure 1: Spatial and temporal verification against independent reference solutions. Errors decrease approximately quadratically in mesh width for the passive cable and in time step for the uniform active membrane. The full-step initialization curve shows the error caused by advancing the initial gates through a full time step instead of placing them at $t = \Delta t/2$.

Advancing the initial gates by a full time step instead gives observed temporal orders of 0.939, 0.969, and 0.985 in this non-equilibrium test. The comparison identifies initialization as a material part of the scheme. These results support second-order behavior for the specified smooth problems; they do not constitute a general convergence proof for heterogeneous branched cables or establish accuracy near a propagation threshold.

5 Propagation through a bifurcation

Consider an active parent cable connected to two active daughters, with no added junction or soma area. Each cable is 10 cm long, the parent radius is $r_p = 0.0238$ cm, and the daughters have equal radii. Define the diameter ratio

$$Q = \frac{d_1^{3/2} + d_2^{3/2}}{d_p^{3/2}}, \qquad r_1 = r_2 = r_p(Q/2)^{2/3}. \tag{12}$$

The distal daughter endpoints are sealed. At the proximal parent endpoint, a total current pulse is applied:

$$I(t) = I_{\mathrm{pk}} \sin^2\!\left(\frac{\pi(t - t_0)}{T}\right), \quad t_0 \le t \le t_0 + T, \tag{13}$$

with $I(t) = 0$ otherwise, $t_0 = 1$ ms, and $T = 2$ ms. The baseline mesh has 100 intervals per cable and $\Delta t = 0.01$ ms. The simulation window is 35 ms. Arrival is defined operationally as the first upward crossing of 0 mV, linearly interpolated between saved samples.

At $Q = 1$, peak currents of 0.1, 0.5, and 1 μA produce no 0-mV crossing at the monitored input, junction, or daughter endpoints within the observation window. A 3-μA pulse produces crossings at approximately 2.355 ms at the input, 10.390 ms at the junction, and 20.473 ms at both daughter endpoints. This sweep brackets a stimulus transition in the tested settings without locating a precise excitation threshold.

All five tested ratios propagate to both daughter endpoints under this stimulus. Thus, $Q > 1$ alone does not imply propagation failure in these examples. Increasing daughter radius also changes conduction along the daughters, so this sweep does not isolate branch-point load. The diameter ratio is a useful geometric descriptor, but active propagation depends on the complete membrane, geometry, and stimulus specification [1].

At $Q = 2$ and 100 intervals per cable, daughter-end arrival changes from 18.470 to 18.462 to 18.459 ms as Δt decreases from 0.02 to 0.01 to 0.005 ms. At fixed $\Delta t = 0.005$ ms, meshes of 50, 100, and 200 intervals give 18.609, 18.459, and 18.421 ms. Spatial sensitivity is therefore larger than the final temporal change in this comparison. These interpolated estimates do not establish accuracy to 0.001 ms. Saved samples are spaced by 0.05 ms, except for 0.06 ms in the 0.02-ms run; interpolation adds another timing uncertainty.

6 Propagation direction and stimulus order

6.1 Matched endpoint pulses

A uniform active cable provides a symmetry control. The cable is 10 cm long with radius 0.0238 cm, and a pulse with $I_{\mathrm{pk}} = 3$ μA, $t_0 = 1$ ms, and $T = 2$ ms is applied at either endpoint. Each pulse delivers total charge $I_{\mathrm{pk}}T/2 = 3$ μA ms. With no additional endpoint area, the full voltage traces agree under spatial reflection to numerical precision. At 400 intervals and $\Delta t = 0.005$ ms, the first upward 0-mV crossing at the opposite endpoint occurs at approximately 10.160 ms in both directions.

Adding 0.01 cm^2 of active membrane area at the root breaks this geometric symmetry. Opposite-end arrival is approximately 10.211 ms for a wave traveling toward the root and 10.222 ms for a wave traveling away. The extra area changes both the electrical load and, when the root is stimulated, the membrane area receiving the injected current. Equal total current consequently does not imply equal current density. The added area is a synthetic lumped load, not a calibrated soma.

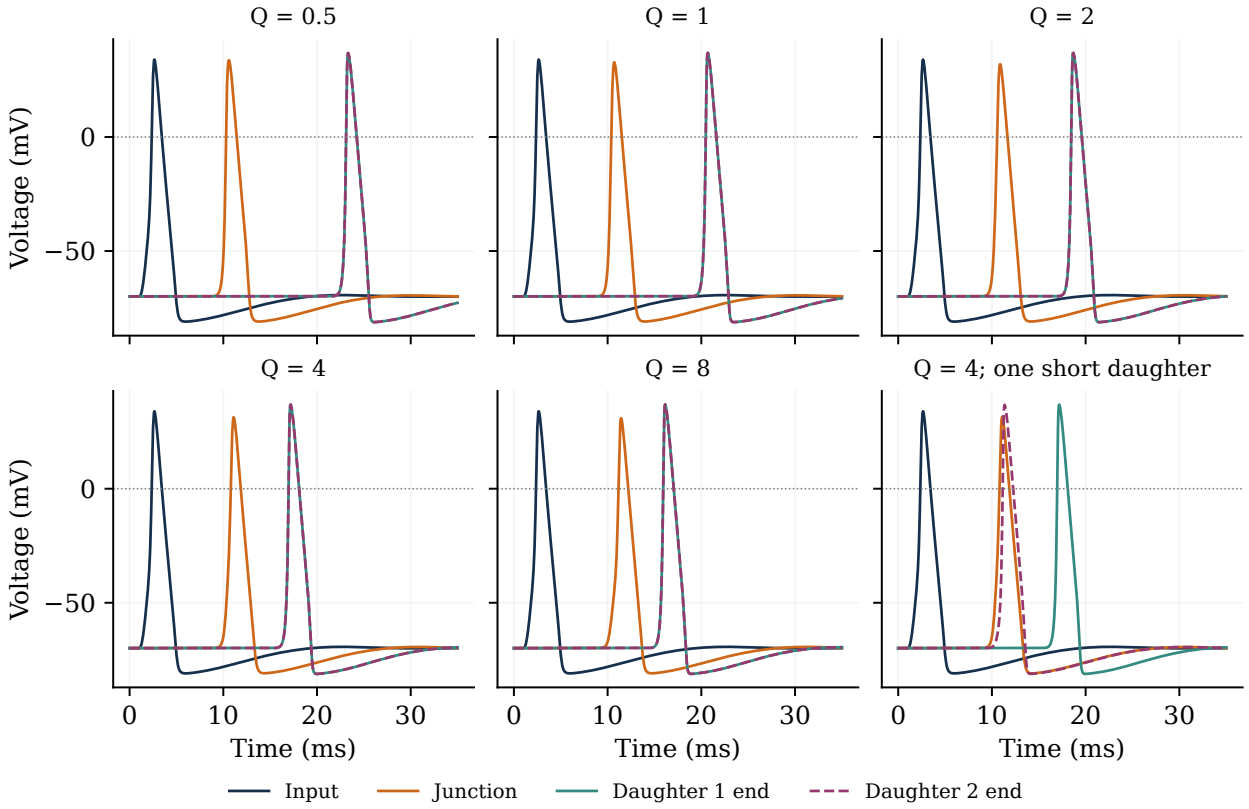


Figure 2: Voltage traces at the parent input, junction, and daughter endpoints under a 3-μA pulse. Both daughter endpoints cross 0 mV for each tested ratio $Q = 0.5, 1, 2, 4, 8$. Equal daughters have coincident traces. The final panel holds $Q = 4$ and shortens one daughter from 10 to 1 cm; both endpoints still cross threshold, at different times.

6.2 Reversing a sequence of local inputs

A second experiment stimulates the sites $x = 2, 4, 6, 8$ cm with four sine-squared pulses of 0.5-ms duration and 0.8-ms onset spacing, beginning at 1 ms. Reversing their order produces sequences traveling toward or away from the root while preserving the sites and total prescribed charge. With 400 intervals, $\Delta t = 0.005$ ms, the root-area load, and 3-μA pulses, the root peak is approximately -69.522 mV for the rootward sequence and -69.542 mV for the outward sequence. Neither crosses 0 mV within 18 ms. The tested passive and lower-amplitude active cases are also subthreshold at the root.

The last spatial refinement changes the reported peaks by less than 3×10^{-5} mV in the tested cases. Nevertheless, reversing the sequence also changes the onset time of the nearest stimulus. Differences in peak timing alone therefore cannot identify an intrinsic directional preference. These experiments illustrate the need to control both geometry and stimulus timing when interpreting direction-dependent responses.

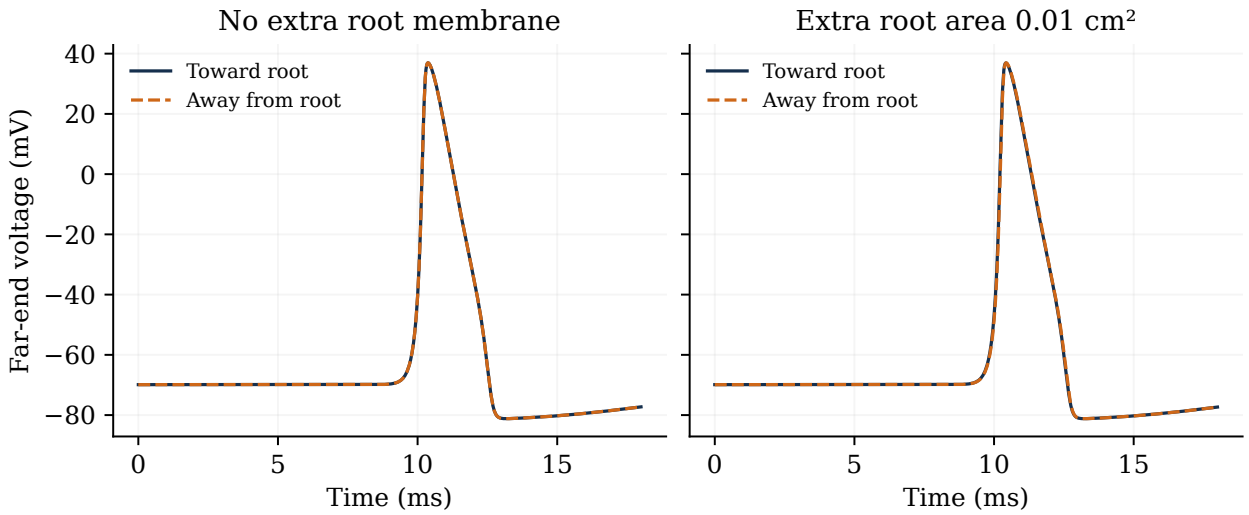


Figure 3: Far-end voltage under matched endpoint pulses. Without extra root membrane, the two directions agree under reflection. An additional root area changes the response. Curves use 400 intervals, $\Delta t = 0.005$ ms, and a saved sample spacing of 0.025 ms. The small arrival-time difference is a model output and does not establish biological directional selectivity.

7 Discussion and future work

The numerical framework separates three questions: whether the discrete equations conserve current, whether a calculation is resolved, and whether its parameters represent a biological neuron. Half-segment membrane areas and a shared junction voltage address conservation. Independent solutions, residuals, gate checks, and refinement address numerical behavior. Physiological interpretation requires additional evidence about morphology, channels, and stimulation.

The examples show successful propagation across a range of daughter diameters and reflection symmetry on a homogeneous cable. They do not establish a universal success or failure boundary for branching. A systematic threshold study would need finer stimulus and geometry sweeps, explicit stopping criteria, and refinement near marginal propagation. Similarly, small directional timing differences should be tested against both temporal and spatial error before being assigned biological meaning.

A useful next project is to connect cable models through stochastic synapses. Work on stochastic vesicle release provides a basis for examining how release variability affects signal transmission [5, 6]. A finite-site vesicle model can represent docking, undocking, and probabilistic release at presynaptic spikes [6]. Release events would then drive a postsynaptic conductance with an explicitly defined time course and reversal potential. This would make it possible to study how release variability interacts with propagation and integration, rather than treating each neuron as an isolated tree.

Measured neuronal morphologies could supply branch connectivity, lengths, and diameters. These geometric data would constrain the cable representation, while membrane channel densities, kinetics, synaptic parameters, and electrophysiological responses would require independent calibration and validation. A staged application could begin with an individual Purkinje-cell model, then connect it to a small network of other neuron models through specified synapses. Stochastic synaptic coupling, morphology-based calibration, and the connected-network simulations are proposed extensions, not results demonstrated here.

8 Conclusion

Conservative membrane-area accounting and a linear-time tree solve make branched Hodgkin–Huxley simulations efficient and testable. The benchmark and propagation results support using this framework for controlled numerical studies, with measured morphology and calibrated stochastic synapses as the next steps toward cell-specific and network models.

Reproducibility

The calculations reported here are supported by saved experiment configurations, voltage and gate histories, numerical diagnostics, and refinement summaries. The computational environment used for the saved results is Python 3.10.15, NumPy 1.26.4, SciPy 1.12.0, and Matplotlib 3.9.2. Figures in this report are plotted from those saved results. No experimental recordings or newly calibrated neuronal parameters are introduced.

Acknowledgments

I thank Charles S. Peskin for his mentorship. The cable discretization and staggered numerical scheme follow his neurophysiology notes [4]; the tree solution is placed in the context of established methods for branched nerve equations [2].

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